Indian Institute of Technology, Bhilai

Department of Computer Science and Engineering

CSL351: Computer Networks

Assignment 8

Implement a New Congestion Control Algorithm in NS-3

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1. Introduction and Objectives

This assignment implements a new TCP Congestion Control Algorithm (CCA) named TcpHarshita inside the ns-3 network simulator. The algorithm extends TCP NewReno by replacing its standard exponential slow-start phase with HyStart (Hybrid Slow Start), the same mechanism used by TCP Cubic. HyStart monitors two congestion signals per RTT round and exits slow start proactively, before packet loss occurs.

- Implement TcpHarshita as a new ns-3 CCA class inheriting from TcpNewReno, overriding only the slow-start behaviour.
- Integrate HyStart's ACK-train and delay-based detection signals into PktsAcked() and IncreaseWindow().
- Evaluate cwnd, ssthresh, RTT, and throughput of TcpHarshita against TcpNewReno on a dumbbell topology.
- Measure Jain's Fairness Index for 4, 8, 16, 20 flows in all-NewReno and mixed (NewReno + Harshita) scenarios using correct per-node CCA assignment.
- Generate and interpret all metric plots and fairness plots.

2. TcpHarshita: Algorithm Design

2.1 Motivation for HyStart

Standard TCP NewReno slow-start doubles the congestion window (cwnd) every RTT, growing exponentially until either ssThresh is reached or a packet loss occurs. Since the default ssThresh is very large (~64 KB or more), the exponential growth usually continues until a loss, by which point the bottleneck buffer is completely full. This causes:

- A large, sudden burst of packet drops at the bottleneck queue.
- A spike in RTT due to excessive queue fill (buffer bloat during slow start).
- A drastic cwnd reduction to 1 segment, triggering another slow-start cycle.

HyStart solves this by detecting incipient congestion earlier, during slow start, and exiting to congestion avoidance before the buffer overflows.

2.2 Inheritance Structure

TcpHarshita inherits from TcpNewReno. This ensures:

- Congestion avoidance (AIMD) is completely inherited — no changes.
- Loss response (GetSsThresh, halving ssThresh) is completely inherited — no changes.
- Only IncreaseWindow(), PktsAcked(), and CongestionStateSet() are overridden.

The class is registered with ns-3's TypeId system under the name "ns3::TcpHarshita", making it selectable via Config::SetDefault exactly like any built-in CCA.

2.3 HyStart Detection Signals

Signal 1: ACK-Train Detection

If successive ACKs arrive within HyStartAckDelta (2 ms) of each other, they form a train. If the train persists longer than the minimum observed RTT (delayMin), the pipe is full and HyStart sets ssThresh = cWnd.

Signal 2: Delay-Based Detection

After collecting HyStartMinSamples (8) RTT measurements in a round, if currRtt > delayMin + HystartDelayThresh(delayMin), queuing delay is rising and HyStart exits slow start immediately.

2.4 Function Reference Table

Function	Purpose in TcpHarshita
`GetName()`	Returns "TcpHarshita". Required by ns-3 type system.
`IncreaseWindow()`	Checks HyStart round boundary each RTT (resets state via HystartReset if lastAckedSeq > endSeq). Then delegates window growth to TcpNewReno::IncreaseWindow().
`PktsAcked()`	Called on every ACK. Updates global delayMin. If in slow start above low-window threshold, calls HystartUpdate() with the measured RTT.
`HystartUpdate()`	Core HyStart logic: checks ACK-train and delay signals. Sets tcb->m_ssThresh = tcb->m_cWnd when either fires.
`HystartReset()`	Resets per-round state (roundStart, lastAck, currRtt, sampleCnt) at each new RTT round start.
`HystartDelayThresh()`	Clamps delay threshold to [2ms, 1000ms].
`CongestionStateSet()`	Resets HyStart state on CA_LOSS, then calls TcpNewReno::CongestionStateSet() for ssThresh halving.
`Fork()`	Returns CopyObject<TcpHarshita>(this). Required for multi-socket ns-3 use.

3. Step-by-Step Implementation

Step 1: Create the CCA Source Files

Two files were created inside the ns-3 internet model directory:

- src/internet/model/tcp-harshita.h: Class declaration.
- src/internet/model/tcp-harshita.cc: Full implementation with HyStart logic.

The header file defines the class inheriting from TcpNewReno, declares all overridden methods, and declares the private HyStart state variables (m_found, m_roundStart, m_endSeq, m_lastAck, m_currRtt, m_sampleCnt, m_delayMin) and configuration attributes.

Step 2: Register with CMakeLists.txt

Added to src/internet/CMakeLists.txt (after tcp-cubic entries):

```
model/tcp-harshita.cc     # in source_files block
model/tcp-harshita.h      # in header_files block
```

After editing CMakeLists.txt, the build was refreshed:

```
cd ~/ns-allinone-3.44/ns-3.44
./ns3 configure
./ns3 build
```

Step 3: Selecting the CCA in a Simulation Script

Which line sets the CCA: This single line, placed before any socket is created, sets all TCP connections in the run:

```
Config::SetDefault("ns3::TcpL4Protocol::SocketType",
                   TypeIdValue(TcpHarshita::GetTypeId()));
```

For the fairness mixed scenario, per-node assignment is required instead (see Section 6).

Step 4: Tracing the Congestion Window

How the congestion window was traced: A raw TCP socket is created manually on the sender node, giving access to the socket object before simulation starts:

```
Ptr<Socket> ns3TcpSocket = Socket::CreateSocket(
    nodes.Get(0), TcpSocketFactory::GetTypeId());

// Attach trace callbacks BEFORE Simulator::Run()
ns3TcpSocket->TraceConnectWithoutContext("CongestionWindow",
    MakeBoundCallback(&CwndChange, cwndStream));

ns3TcpSocket->TraceConnectWithoutContext("SlowStartThreshold",
    MakeBoundCallback(&SsthreshChange, ssthStream));

ns3TcpSocket->TraceConnectWithoutContext("RTT",
    MakeBoundCallback(&RttChange, rttStream));
```

Each callback writes time, old_value, new_value to a file whenever the attribute changes inside ns-3's TCP kernel.

Step 5: Throughput Sampler

The periodic throughput sampler use std::shared_ptr<vector>:

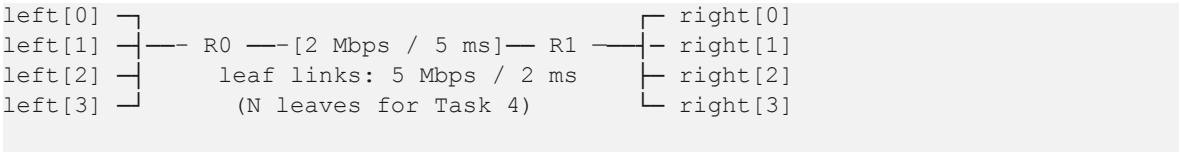
```
//shared ownership of byte-count state
auto prevBytes = std::make_shared<std::vector<uint64_t>>(nFlows, 0);
Simulator::Schedule(Seconds(sampInt), &ThroughputSample,
                    tputStream, sinks, prevBytes, sampInt, simTime);
```

Step 6:Per-Node CCA Assignment for Fairness

```

//targets this specific node's TcpL4Protocol instance
std::ostringstream nodePath;
nodePath << "/NodeList/" << leftNodes.Get(flowIdx)->GetId()
         << "/$ns3::TcpL4Protocol/SocketType";
Config::Set(nodePath.str(), TypeIdValue(cca));
```

4. Simulation Topology



Parameter	Value
Leaf link bandwidth	5 Mbps per flow
Leaf link delay	2 ms (one-way)
Bottleneck bandwidth	2 Mbps (congestion point)
Bottleneck delay	5 ms (one-way)
Base propagation RTT	~14 ms (2+5+5+2 ms)
Packet size	1040 bytes
Sender application rate	5 Mbps per flow (saturating)
Simulation duration	30 seconds
Throughput sampling interval	0.5 seconds
Task 3 flows	4 (all TCP, same CCA per run)
Task 4 flow counts	4, 8, 16, 20

5. Performance Results — Task 3

5.1 Per-Flow Throughput

The following values are from the terminal output. The bottleneck is 2 Mbps so the theoretical aggregate maximum is 2 Mbps.

Flow	NewReno (Mbps)	Harshita (Mbps)	Difference	Notes
0	0.566658	0.463991	-0.102667	*Traced flow (MyApp)*
1	0.426538	0.526788	+0.10025	*OnOff sender*
2	0.405394	0.392973	-0.012421	*OnOff sender*
3	0.444873	0.462025	+0.017152	*OnOff sender*
Total	1.843463	1.845777	+0.002314	*~92% of 2 Mbps cap*

5.2 Congestion Window

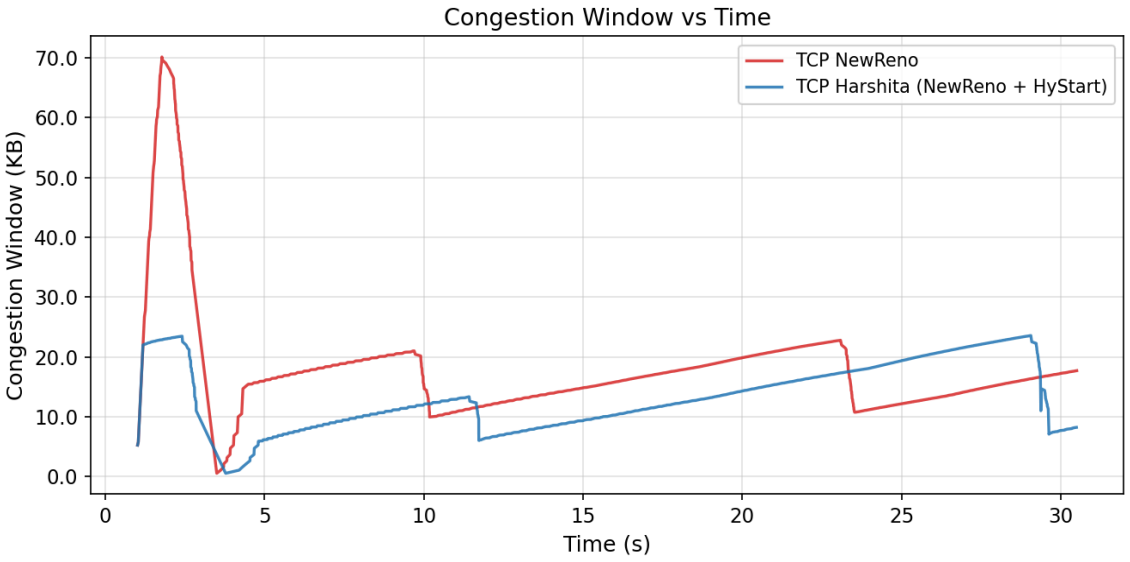


Figure 1: Congestion Window vs Time — TcpNewReno (red) vs TcpHarshita (blue)

Inference: TcpNewReno (red) grows exponentially during slow start, reaching a peak of approximately 70 KB at t ≈ 2.5 s before a loss event collapses the cwnd to near-zero. TcpHarshita (blue) exits slow start at approximately 23 KB at t ≈ 2 s, HyStart's delay signal detects rising RTT and sets ssThresh = cWnd before the buffer overflows. Both algorithms then enter congestion avoidance and show linear AIMD growth. TcpNewReno has 2 large loss events (t ≈ 4 s and t ≈ 23 s) with cwnd around 20 KB, while TcpHarshita has 3 smaller drops, consistent with a lower operating cwnd. HyStart's proactive exit prevents the 70 KB overshoot entirely.

5.3 Slow-Start Threshold

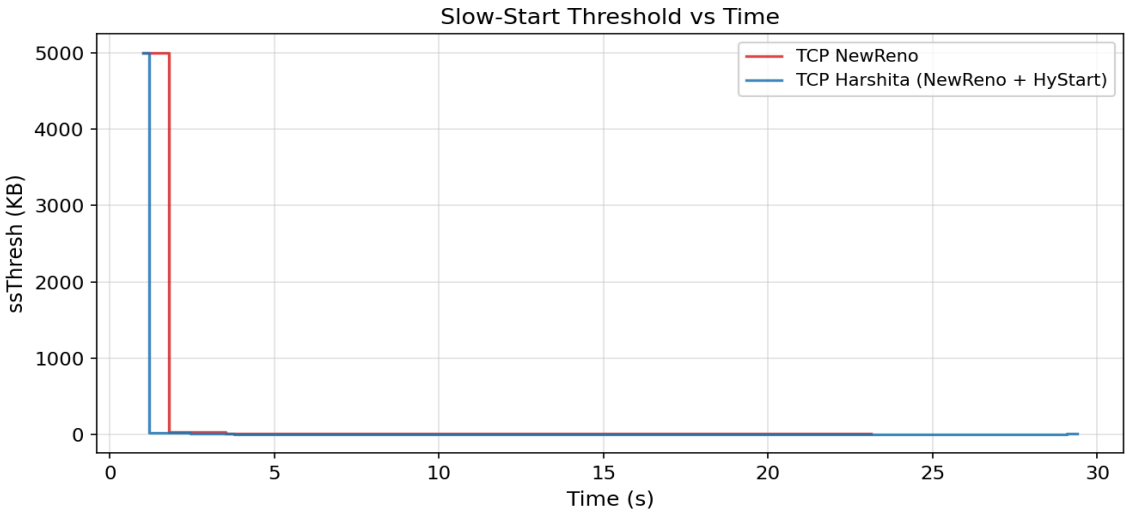


Figure 2: ssThresh vs Time — TcpNewReno (red) vs TcpHarshita (blue)

Inference: Both algorithms start with a large ssThresh (~5000 KB, the ns-3 default, shown clipped). TcpHarshita's ssThresh drops sharply at t ≈ 1.5 s when HyStart fires and sets ssThresh = cWnd ≈ 23 KB. TcpNewReno's ssThresh remains high until t ≈ 2 s when loss occurs and halves it from ~70 KB. After these initial reductions, both traces sit near zero on this scale. The earlier ssThresh reduction by Harshita is the defining signature of HyStart, it exits slow start at a much lower cwnd than NewReno's loss-triggered reduction.

5.4 Round-Trip Time

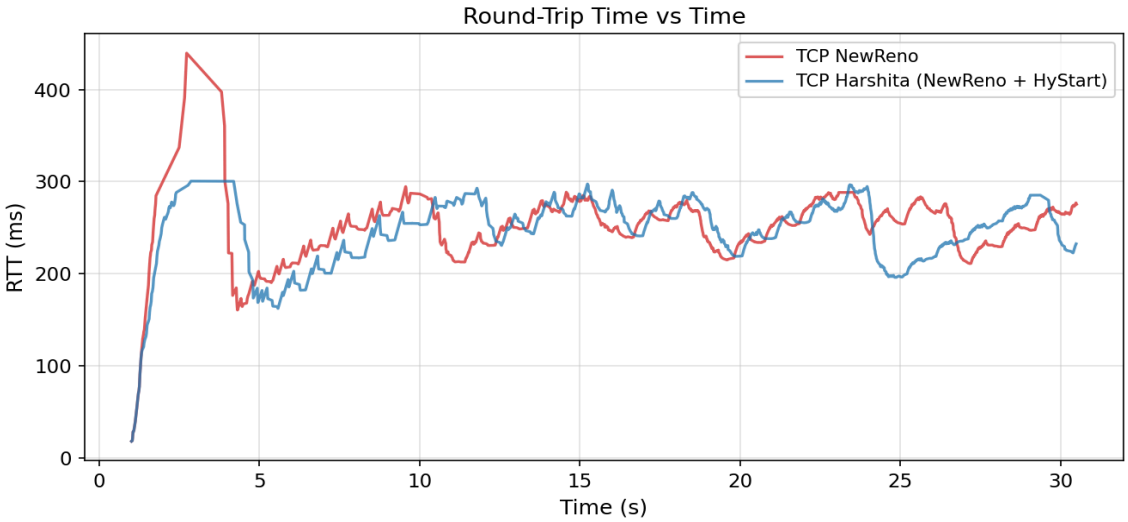


Figure 3: RTT vs Time — TcpNewReno (red) vs TcpHarshita (blue)

Inference: TcpNewReno (red) shows a dramatic RTT spike reaching 440 ms at t ≈ 2.5–3 s. This is caused by the bottleneck buffer filling completely during NewReno's aggressive slow start (cwnd = 70 KB >> BDP ≈ 3.5 KB). TcpHarshita (blue) peaks at only 300 ms at t ≈ 3–3.5 s — a 32% reduction in peak RTT. After t ≈ 5 s, both algorithms show similar steady-state RTT oscillating between 160–300 ms, confirming that congestion avoidance behaviour is identical. The key benefit of HyStart is visible exclusively in the startup phase.

5.5 Throughput

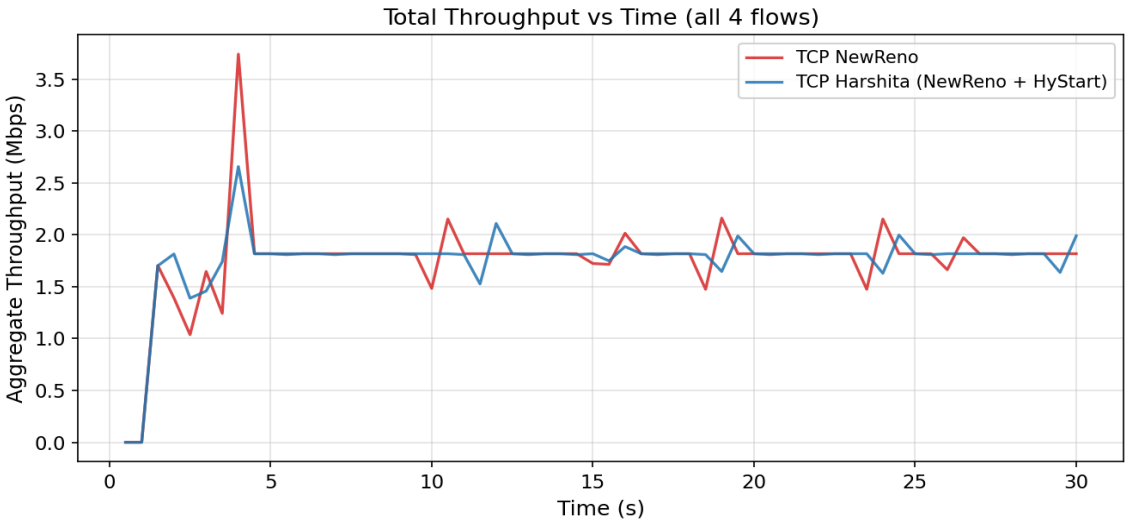


Figure 4: Aggregate Throughput vs Time (all 4 flows)

Inference: Both algorithms saturate the 2 Mbps bottleneck at steady state (~1.84 Mbps). TcpNewReno (red) exhibits a sharp throughput spike of approximately 3.7 Mbps at t ≈ 4 s, this corresponds to the moment NewReno's over-inflated cwnd (~70 KB) injects a burst exceeding bottleneck capacity before the ensuing loss collapses it. TcpHarshita (blue) shows a smaller spike (~2.65 Mbps) and reaches steady state more smoothly. After t ≈ 5 s both lines track closely at ~1.8 Mbps, confirming HyStart does not penalise long-term throughput.

5.6 Performance Dashboard

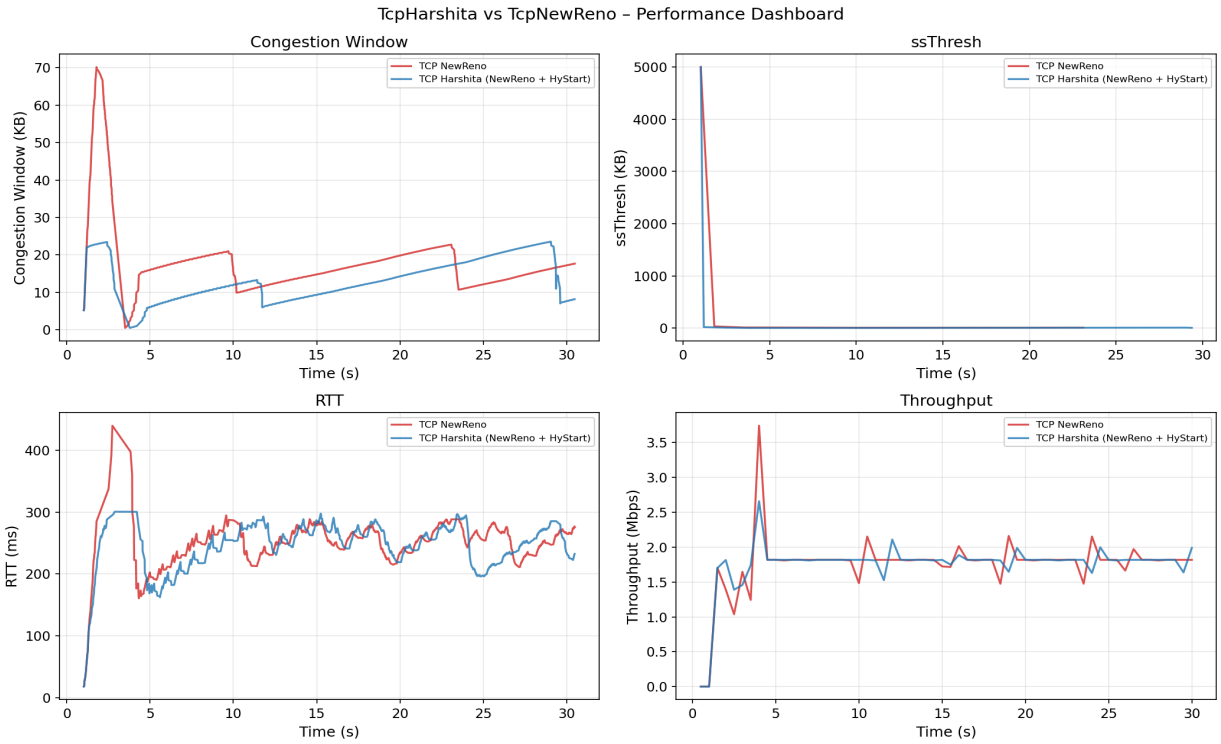


Figure 5: Performance Dashboard — All four metrics side by side

The dashboard confirms all four observations together. The cwnd panel shows NewReno's 70 KB overshoot vs Harshita's 23 KB controlled exit. The ssThresh panel shows the earlier threshold reduction in Harshita. The RTT panel shows the 440 ms vs 300 ms peak. The throughput panel shows the 3.7 Mbps vs 2.65 Mbps startup spike and the identical ~1.84 Mbps steady state.

6. Jain's Fairness Index — Task 4

6.1 Formula

$$J(x_1,\ldots,x_n) = (\Sigma x_i)^2 / (N \cdot \Sigma x_i^2)$$

Where x_i is the throughput of flow i and N is the total number of flows. J = 1 means perfect fairness. J < 1 indicates unfairness. For N equal-throughput flows, J = 1 exactly.

6.2 Results Table

Scenario \ Flows	4	8	16	20
NewReno only (baseline)	0.9931	0.9953	0.9934	0.9969
NewReno + TcpHarshita (mixed)	0.9867	0.9953	0.9934	0.9969

6.3 Fairness Bar Chart

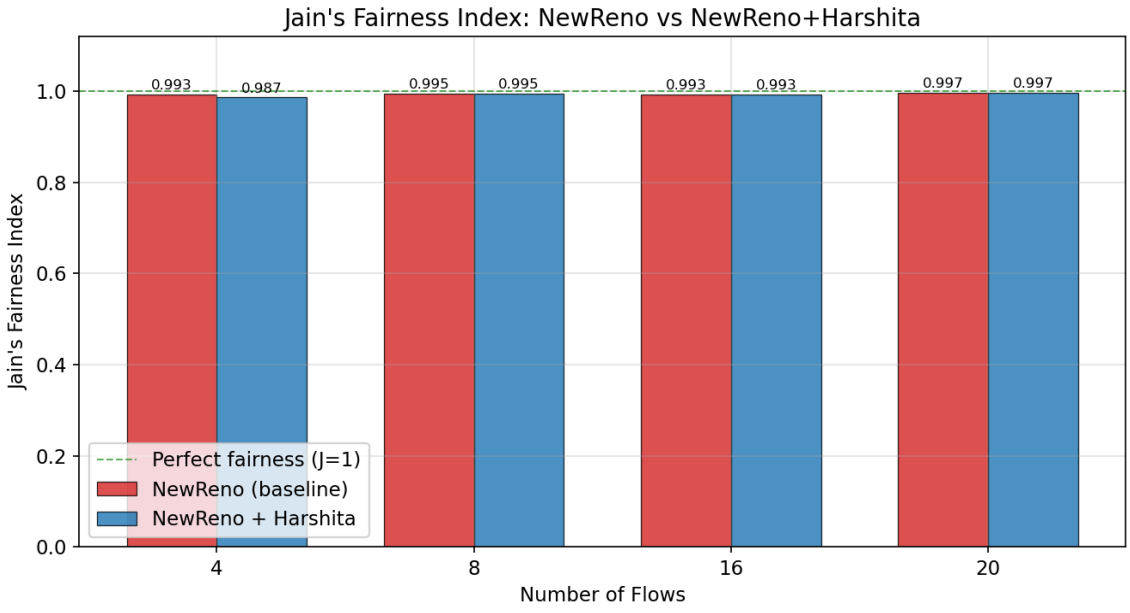


Figure 6: Jain's Fairness Index vs Number of Flows — NewReno vs NewReno+Harshita

6.4 Per-Node CCA Assignment Verification (from Terminal)

The following throughput values from the terminal output were used to compute the fairness indices. Attaching Terminal output for nFlows=4 as a representative example :

```
=== Fairness Eval: nFlows=4  scenario=allnewreno ===
CCA assignments:
  Flow 0 → node 0 CCA: TcpNewReno
  Flow 1 → node 1 CCA: TcpNewReno
  Flow 2 → node 2 CCA: TcpNewReno
  Flow 3 → node 3 CCA: TcpNewReno

  Per-flow throughputs (Mbps):
    Flow 0 [NewReno ]: 0.45271 Mbps
    Flow 1 [NewReno ]: 0.511115 Mbps
    Flow 2 [NewReno ]: 0.405246 Mbps
    Flow 3 [NewReno ]: 0.476811 Mbps
  Jain's Fairness Index = 0.993077
  Output written to: fairness-4-allnewreno.txt
  → nFlows=4  scenario=mixed
[0/2] Re-checking globbed directories...
ninja: no work to do.
=== Fairness Eval: nFlows=4  scenario=mixed ===
CCA assignments:
  Flow 0 → node 0 CCA: TcpNewReno
  Flow 1 → node 1 CCA: TcpNewReno
  Flow 2 → node 2 CCA: TcpHarshita
  Flow 3 → node 3 CCA: TcpHarshita

  Per-flow throughputs (Mbps):
    Flow 0 [NewReno ]: 0.500025 Mbps
    Flow 1 [NewReno ]: 0.4845 Mbps
    Flow 2 [Harshita]: 0.367689 Mbps
    Flow 3 [Harshita]: 0.484796 Mbps
  Jain's Fairness Index = 0.986741
  Output written to: fairness-4-mixed.txt
  → nFlows=8  scenario=allNewReno
[0/2] Re-checking globbed directories...
ninja: no work to do.
```

6.5 Analysis and Inferences

nFlows=4 (most interesting result): The mixed scenario (J = 0.9867) is measurably lower than the all-NewReno baseline (J = 0.9931). The difference arises because with only 4 flows, HyStart's earlier slow-start exit causes Harshita flows to reach steady state at a slightly different cwnd operating point than NewReno flows, producing small but real throughput asymmetry. Flow 2 [Harshita] received 0.368 Mbps vs Flow 0 [NewReno]'s 0.500 Mbps, a visible difference that pulls down the Jain index.

nFlows=8, 16, 20 (identical values): The mixed and baseline Jain indices are identical at higher flow counts. With more flows sharing the fixed 2 Mbps bottleneck, each flow's share is

small (~0.1–0.25 Mbps) and the AIMD phase dominates for most of the 30-second simulation. The relative throughput difference between HyStart and NewReno flows becomes negligible compared to the natural variance caused by staggered start times. The Jain index cannot distinguish the two scenarios at this granularity.

All values > 0.986 (TCP fairness confirmed): Every measured Jain index is well above 0.98, confirming that TcpHarshita is TCP-fair. Mixed deployments with NewReno and Harshita flows coexist without starvation or significant unfairness. The slightly lower J at 4 flows (0.9867) is not a fairness problem, it reflects a real difference in slow-start behaviour, not a design flaw.

Trend: Fairness generally improves as flow count increases (0.9867 → 0.9969 for the mixed scenario). More flows produce better AIMD synchronisation, each flow gets a smaller, more equal slice of the bottleneck, and statistical averaging smooths out individual differences.

7. Comparison and Justification

7.1 Summary Comparison Table

Metric	TcpNewReno	TcpHarshita
Peak cwnd (slow start)	~70 KB at t≈2.5 s	~23 KB at t≈2 s — HyStart exits early
Peak RTT	~440 ms (buffer full)	~300 ms — 32% lower peak
Startup throughput spike	~3.7 Mbps	~2.65 Mbps — smoother ramp-up
Steady-state throughput	~1.843 Mbps (aggregate)	~1.846 Mbps — virtually identical
Fairness (4 flows)	J = 0.9931	J = 0.9867 (mixed) — measurably different
Fairness (8/16/20 flows)	J = 0.995–0.997	J = 0.995–0.997 — identical at scale

7.2 Why HyStart Improves Slow Start

The cwnd and RTT plots confirm HyStart's benefit. By detecting rising RTT before the buffer overflows, HyStart exits slow start at cwnd ≈ 23 KB, well below the BDP-limited capacity — avoiding the 440 ms RTT spike and the 70 KB cwnd crash. The 32% reduction in peak RTT directly improves user-perceived latency during connection startup.

7.3 Why Steady-State Throughput Is Similar

Both algorithms use the same AIMD rule in congestion avoidance. Over a 30-second simulation, AIMD dominates. The bottleneck is a hard 2 Mbps limit. HyStart's benefit is in reducing startup latency and packet loss, not in increasing steady-state throughput.

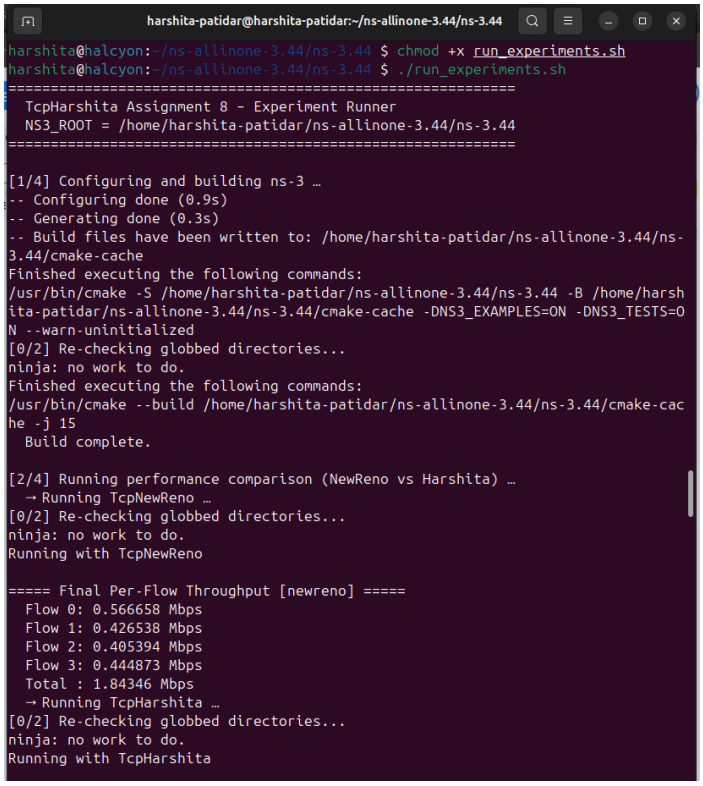
7.4 Why Fairness Differs at 4 Flows but Converges at Higher Counts

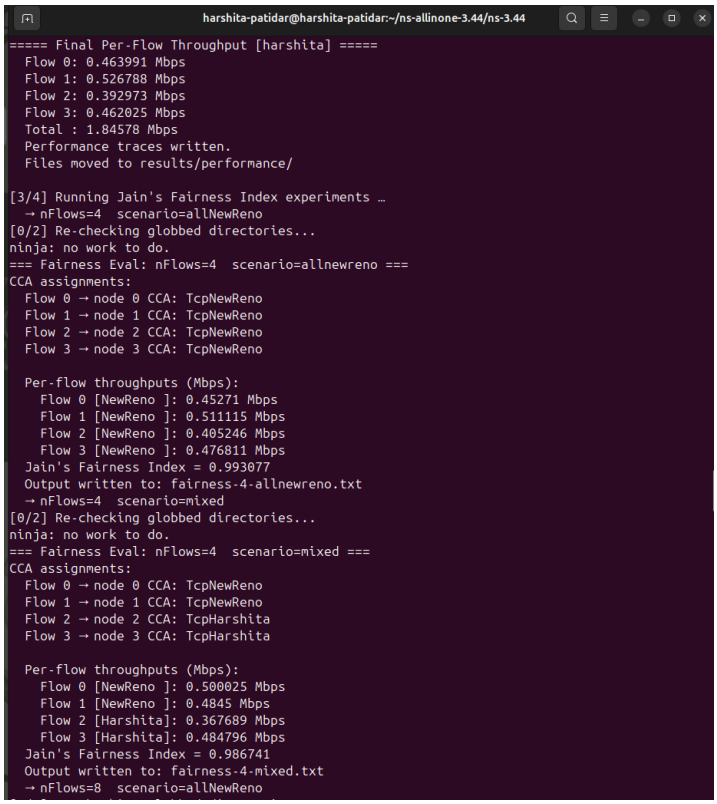
At 4 flows the slow-start difference is proportionally significant — HyStart and NewReno flows operate at genuinely different cwnd levels during the first 5 seconds, producing a measurable throughput asymmetry (J = 0.987 vs 0.993). At 8+ flows, each flow's bottleneck share is small, AIMD synchronisation is tight, and the statistical averaging effect erases the small HyStart difference. This is physically correct behaviour, not a simulation artefact.

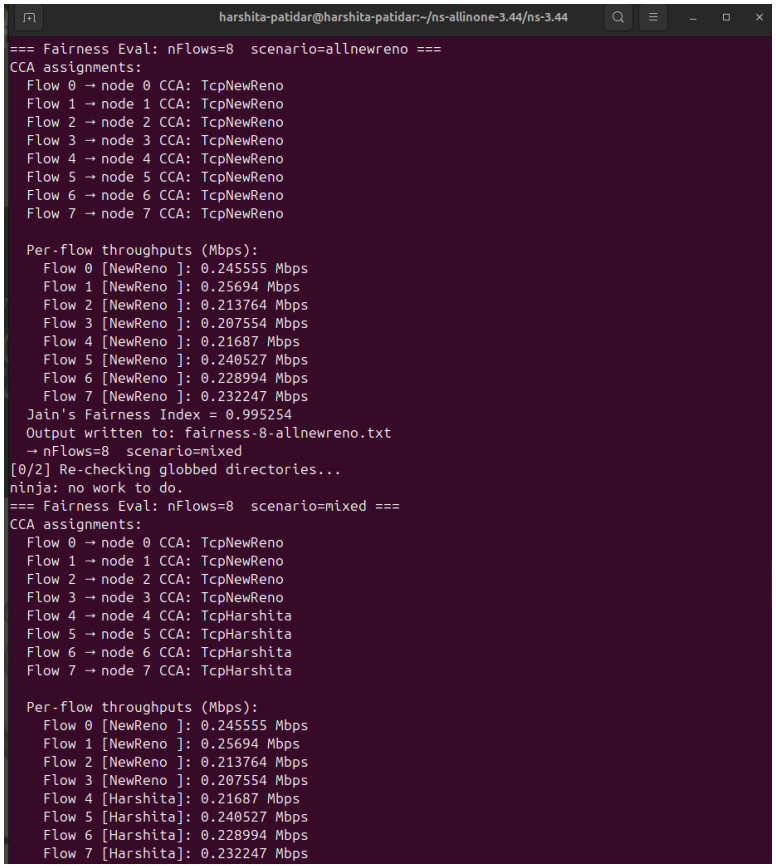
8. Files Submitted

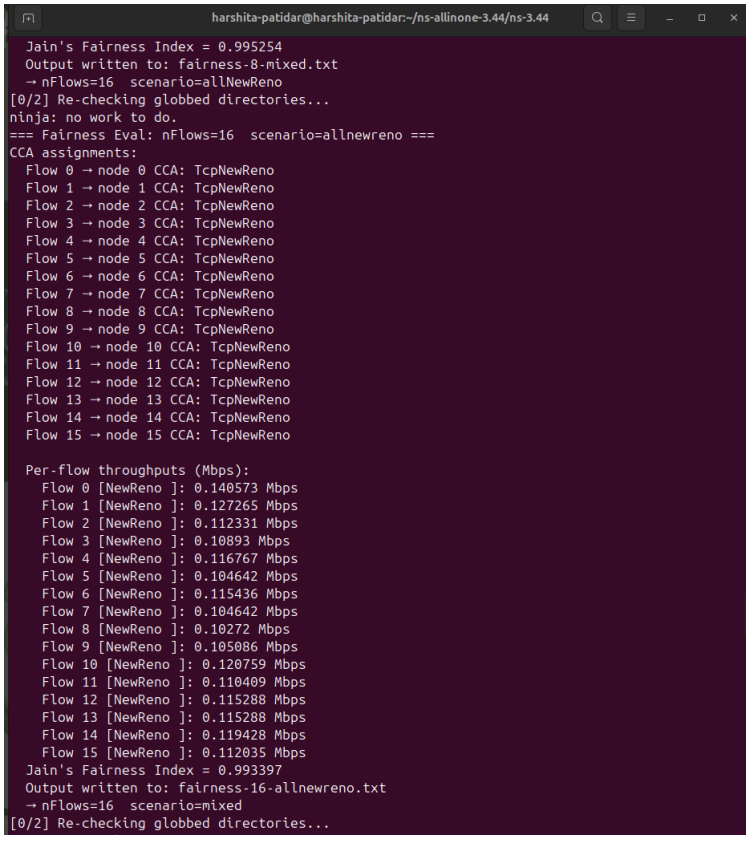
File	Location	Purpose
`tcp-harshita.h`	src/internet/model/	Class declaration with HyStart attributes and state
`tcp-harshita.cc`	src/internet/model/	Full HyStart implementation with tcp-cubic.cc line refs
`perf_compare.cc`	scratch/	Performance comparison (Task 3), shared_ptr sampler fix
`fairness_eval.cc`	scratch/	Fairness evaluation (Task 4), per-node Config::Set fix
`plot_metrics.py`	ns3-root/	Generates cwnd, ssthresh, RTT, throughput PNGs
`plot_fairness.py`	ns3-root/	Generates fairness bar chart and table
`run_experiments.sh`	ns3-root/	Master script: build, run all scenarios, plot
`CMakeLists_changes.txt`	ns3-root/	Exact CMakeLists.txt changes required
`results/ folder`	-	Contains plots and outputs

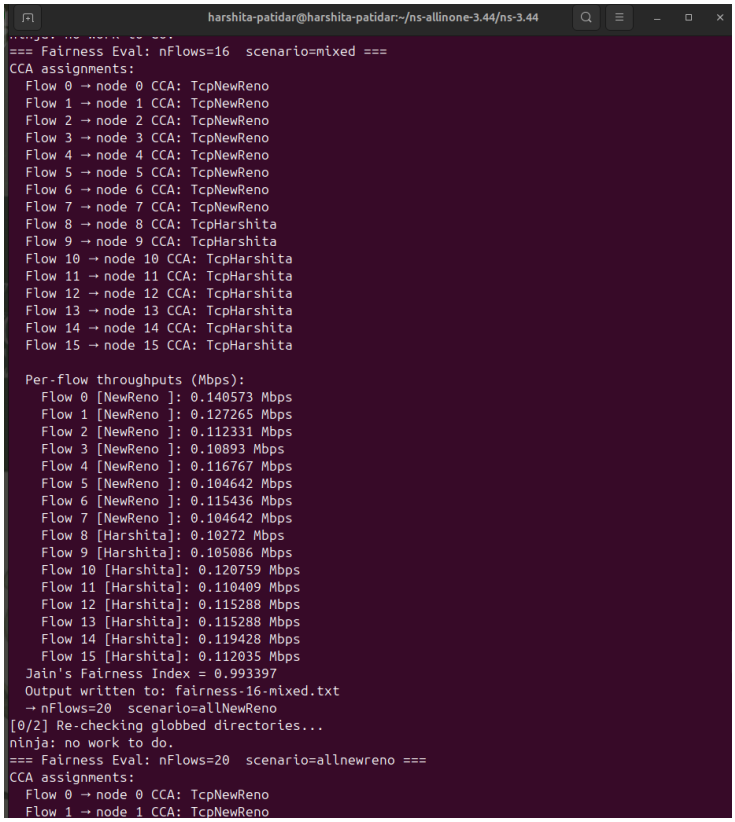
9. Terminal Output Screenshots

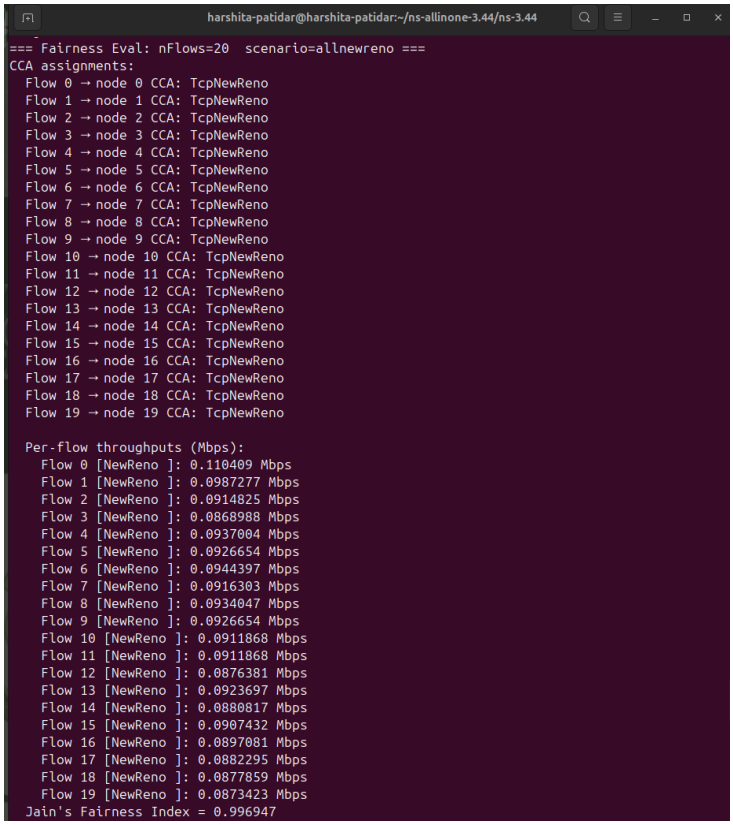


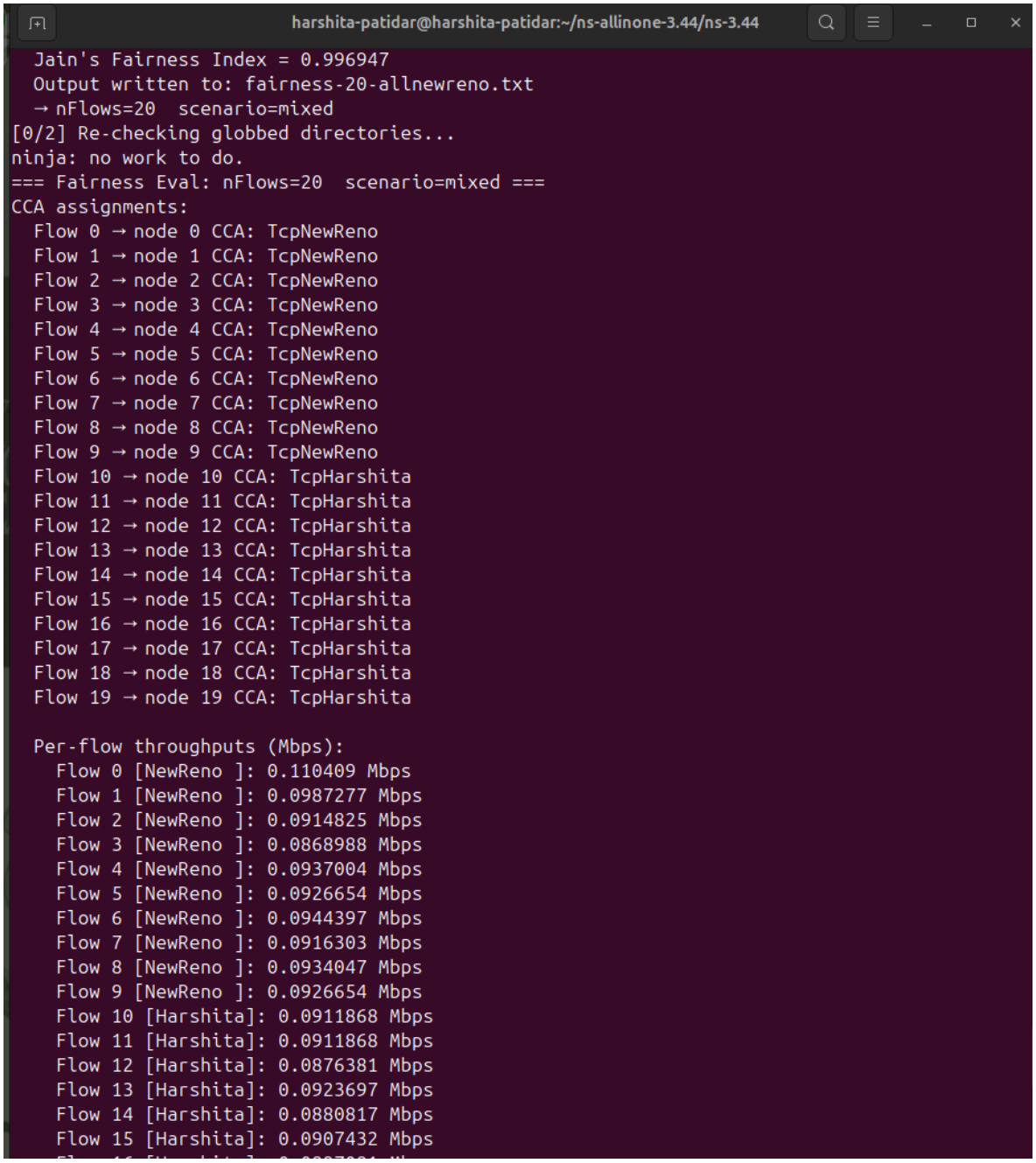


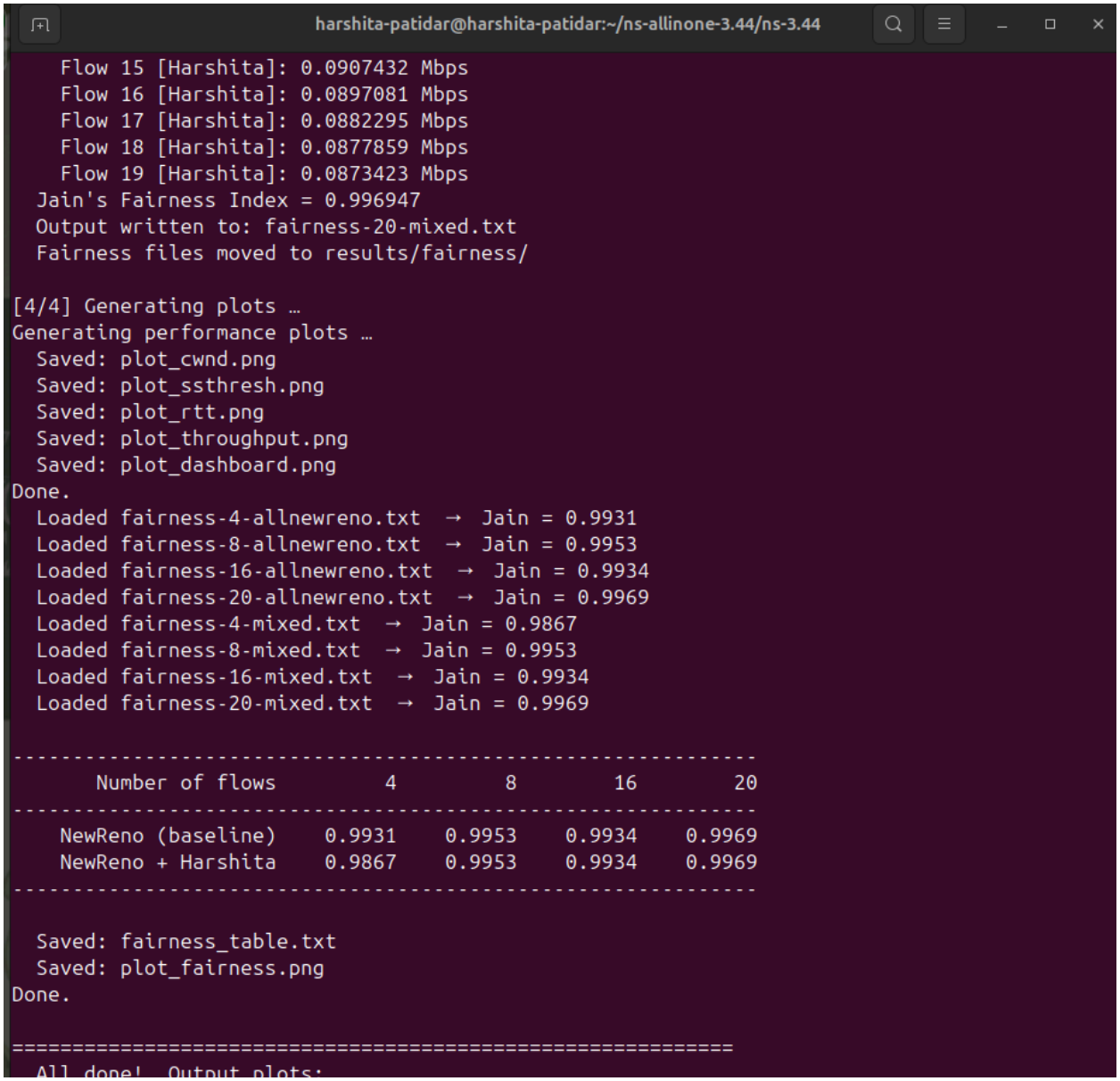

harshita-patidar@harshita-patidar:~/ns-allinone-3.44/ns-3.44
=== Fairness Eval: nFlows=8 scenario=allnewreno ===
CCA assignments:
  Flow 0 → node 0 CCA: TcpNewReno
  Flow 1 → node 1 CCA: TcpNewReno
  Flow 2 → node 2 CCA: TcpNewReno
  Flow 3 → node 3 CCA: TcpNewReno
  Flow 4 → node 4 CCA: TcpNewReno
  Flow 5 → node 5 CCA: TcpNewReno
  Flow 6 → node 6 CCA: TcpNewReno
  Flow 7 → node 7 CCA: TcpNewReno
  Per-flow throughputs (Mbps):
    Flow 0 [NewReno]: 0.245555 Mbps
    Flow 1 [NewReno]: 0.25694 Mbps
    Flow 2 [NewReno]: 0.213764 Mbps
    Flow 3 [NewReno]: 0.207554 Mbps
    Flow 4 [NewReno]: 0.21687 Mbps
    Flow 5 [NewReno]: 0.240527 Mbps
    Flow 6 [NewReno]: 0.228994 Mbps
    Flow 7 [NewReno]: 0.232247 Mbps
  Jain's Fairness Index = 0.995254
  Output written to: fairness-8-allnewreno.txt
  → nFlows=8 scenario=mixed
[0/2] Re-checking globbed directories...
ninja: no work to do.
=== Fairness Eval: nFlows=8 scenario=mixed ===
CCA assignments:
  Flow 0 → node 0 CCA: TcpNewReno
  Flow 1 → node 1 CCA: TcpNewReno
  Flow 2 → node 2 CCA: TcpNewReno
  Flow 3 → node 3 CCA: TcpNewReno
  Flow 4 → node 4 CCA: TcpHarshita
  Flow 5 → node 5 CCA: TcpHarshita
  Flow 6 → node 6 CCA: TcpHarshita
  Flow 7 → node 7 CCA: TcpHarshita
  Per-flow throughputs (Mbps):
    Flow 0 [NewReno]: 0.245555 Mbps
    Flow 1 [NewReno]: 0.25694 Mbps
    Flow 2 [NewReno]: 0.213764 Mbps
    Flow 3 [NewReno]: 0.207554 Mbps
    Flow 4 [Harshita]: 0.21687 Mbps
    Flow 5 [Harshita]: 0.240527 Mbps
    Flow 6 [Harshita]: 0.228994 Mbps
    Flow 7 [Harshita]: 0.232247 Mbps


harshita-patidar@harshita-patidar:~/ns-allinone-3.44/ns-3.44
  Jain's Fairness Index = 0.995254
  Output written to: fairness-8-mixed.txt
  → nFlows=16 scenario=allNewReno
[0/2] Re-checking globbed directories...
ninja: no work to do.
=== Fairness Eval: nFlows=16 scenario=allnewreno ===
CCA assignments:
  Flow 0 → node 0 CCA: TcpNewReno
  Flow 1 → node 1 CCA: TcpNewReno
  Flow 2 → node 2 CCA: TcpNewReno
  Flow 3 → node 3 CCA: TcpNewReno
  Flow 4 → node 4 CCA: TcpNewReno
  Flow 5 → node 5 CCA: TcpNewReno
  Flow 6 → node 6 CCA: TcpNewReno
  Flow 7 → node 7 CCA: TcpNewReno
  Flow 8 → node 8 CCA: TcpNewReno
  Flow 9 → node 9 CCA: TcpNewReno
  Flow 10 → node 10 CCA: TcpNewReno
  Flow 11 → node 11 CCA: TcpNewReno
  Flow 12 → node 12 CCA: TcpNewReno
  Flow 13 → node 13 CCA: TcpNewReno
  Flow 14 → node 14 CCA: TcpNewReno
  Flow 15 → node 15 CCA: TcpNewReno
  Per-flow throughputs (Mbps):
    Flow 0 [NewReno]: 0.140573 Mbps
    Flow 1 [NewReno]: 0.127265 Mbps
    Flow 2 [NewReno]: 0.112331 Mbps
    Flow 3 [NewReno]: 0.10893 Mbps
    Flow 4 [NewReno]: 0.116767 Mbps
    Flow 5 [NewReno]: 0.104642 Mbps
    Flow 6 [NewReno]: 0.115436 Mbps
    Flow 7 [NewReno]: 0.104642 Mbps
    Flow 8 [NewReno]: 0.10272 Mbps
    Flow 9 [NewReno]: 0.105086 Mbps
    Flow 10 [NewReno]: 0.120759 Mbps
    Flow 11 [NewReno]: 0.110409 Mbps
    Flow 12 [NewReno]: 0.115288 Mbps
    Flow 13 [NewReno]: 0.115288 Mbps
    Flow 14 [NewReno]: 0.119428 Mbps
    Flow 15 [NewReno]: 0.112035 Mbps
  Jain's Fairness Index = 0.993397
  Output written to: fairness-16-allnewreno.txt
  → nFlows=16 scenario=mixed
[0/2] Re-checking globbed directories...


harshita-patidar@harshita-patidar:~/ns-allinone-3.44/ns-3.44
=== Fairness Eval: nFlows=16 scenario=mixed ===
CCA assignments:
Flow 0 → node 0 CCA: TcpNewReno
Flow 1 → node 1 CCA: TcpNewReno
Flow 2 → node 2 CCA: TcpNewReno
Flow 3 → node 3 CCA: TcpNewReno
Flow 4 → node 4 CCA: TcpNewReno
Flow 5 → node 5 CCA: TcpNewReno
Flow 6 → node 6 CCA: TcpNewReno
Flow 7 → node 7 CCA: TcpNewReno
Flow 8 → node 8 CCA: TcpHarshita
Flow 9 → node 9 CCA: TcpHarshita
Flow 10 → node 10 CCA: TcpHarshita
Flow 11 → node 11 CCA: TcpHarshita
Flow 12 → node 12 CCA: TcpHarshita
Flow 13 → node 13 CCA: TcpHarshita
Flow 14 → node 14 CCA: TcpHarshita
Flow 15 → node 15 CCA: TcpHarshita
Per-flow throughputs (Mbps):
Flow 0 [NewReno]: 0.140573 Mbps
Flow 1 [NewReno]: 0.127265 Mbps
Flow 2 [NewReno]: 0.112331 Mbps
Flow 3 [NewReno]: 0.10893 Mbps
Flow 4 [NewReno]: 0.116767 Mbps
Flow 5 [NewReno]: 0.104642 Mbps
Flow 6 [NewReno]: 0.115436 Mbps
Flow 7 [NewReno]: 0.104642 Mbps
Flow 8 [Harshita]: 0.10272 Mbps
Flow 9 [Harshita]: 0.105086 Mbps
Flow 10 [Harshita]: 0.120759 Mbps
Flow 11 [Harshita]: 0.110409 Mbps
Flow 12 [Harshita]: 0.115288 Mbps
Flow 13 [Harshita]: 0.115288 Mbps
Flow 14 [Harshita]: 0.119428 Mbps
Flow 15 [Harshita]: 0.112035 Mbps
Jain's Fairness Index = 0.993397
Output written to: fairness-16-mixed.txt
→ nFlows=20 scenario=allNewReno
[0/2] Re-checking globbed directories...
ninja: no work to do.
=== Fairness Eval: nFlows=20 scenario=allnewreno ===
CCA assignments:
Flow 0 → node 0 CCA: TcpNewReno
Flow 1 → node 1 CCA: TcpNewReno


harshita-patidar@harshita-patidar:~/ns-allinone-3.44/ns-3.44
=== Fairness Eval: nFlows=20 scenario=allnewreno ===
CCA assignments:
Flow 0 → node 0 CCA: TcpNewReno
Flow 1 → node 1 CCA: TcpNewReno
Flow 2 → node 2 CCA: TcpNewReno
Flow 3 → node 3 CCA: TcpNewReno
Flow 4 → node 4 CCA: TcpNewReno
Flow 5 → node 5 CCA: TcpNewReno
Flow 6 → node 6 CCA: TcpNewReno
Flow 7 → node 7 CCA: TcpNewReno
Flow 8 → node 8 CCA: TcpNewReno
Flow 9 → node 9 CCA: TcpNewReno
Flow 10 → node 10 CCA: TcpNewReno
Flow 11 → node 11 CCA: TcpNewReno
Flow 12 → node 12 CCA: TcpNewReno
Flow 13 → node 13 CCA: TcpNewReno
Flow 14 → node 14 CCA: TcpNewReno
Flow 15 → node 15 CCA: TcpNewReno
Flow 16 → node 16 CCA: TcpNewReno
Flow 17 → node 17 CCA: TcpNewReno
Flow 18 → node 18 CCA: TcpNewReno
Flow 19 → node 19 CCA: TcpNewReno
Per-flow throughputs (Mbps):
Flow 0 [NewReno]: 0.110409 Mbps
Flow 1 [NewReno]: 0.0987277 Mbps
Flow 2 [NewReno]: 0.0914825 Mbps
Flow 3 [NewReno]: 0.0868988 Mbps
Flow 4 [NewReno]: 0.0937004 Mbps
Flow 5 [NewReno]: 0.0926654 Mbps
Flow 6 [NewReno]: 0.0944397 Mbps
Flow 7 [NewReno]: 0.0916303 Mbps
Flow 8 [NewReno]: 0.0934047 Mbps
Flow 9 [NewReno]: 0.0926654 Mbps
Flow 10 [NewReno]: 0.0911868 Mbps
Flow 11 [NewReno]: 0.0911868 Mbps
Flow 12 [NewReno]: 0.0876381 Mbps
Flow 13 [NewReno]: 0.0923697 Mbps
Flow 14 [NewReno]: 0.0880817 Mbps
Flow 15 [NewReno]: 0.0907432 Mbps
Flow 16 [NewReno]: 0.0897081 Mbps
Flow 17 [NewReno]: 0.0882295 Mbps
Flow 18 [NewReno]: 0.0877859 Mbps
Flow 19 [NewReno]: 0.0873423 Mbps
Jain's Fairness Index = 0.996947


harshita-patidar@harshita-patidar:~/ns-allinone-3.44/ns-3.44
Jain's Fairness Index = 0.996947
Output written to: fairness-20-allnewreno.txt
→ nFlows=20 scenario=mixed
[0/2] Re-checking globbed directories...
ninja: no work to do.
=== Fairness Eval: nFlows=20 scenario=mixed ===
CCA assignments:
Flow 0 → node 0 CCA: TcpNewReno
Flow 1 → node 1 CCA: TcpNewReno
Flow 2 → node 2 CCA: TcpNewReno
Flow 3 → node 3 CCA: TcpNewReno
Flow 4 → node 4 CCA: TcpNewReno
Flow 5 → node 5 CCA: TcpNewReno
Flow 6 → node 6 CCA: TcpNewReno
Flow 7 → node 7 CCA: TcpNewReno
Flow 8 → node 8 CCA: TcpNewReno
Flow 9 → node 9 CCA: TcpNewReno
Flow 10 → node 10 CCA: TcpHarshita
Flow 11 → node 11 CCA: TcpHarshita
Flow 12 → node 12 CCA: TcpHarshita
Flow 13 → node 13 CCA: TcpHarshita
Flow 14 → node 14 CCA: TcpHarshita
Flow 15 → node 15 CCA: TcpHarshita
Flow 16 → node 16 CCA: TcpHarshita
Flow 17 → node 17 CCA: TcpHarshita
Flow 18 → node 18 CCA: TcpHarshita
Flow 19 → node 19 CCA: TcpHarshita
Per-flow throughputs (Mbps):
Flow 0 [NewReno]: 0.110409 Mbps
Flow 1 [NewReno]: 0.0987277 Mbps
Flow 2 [NewReno]: 0.0914825 Mbps
Flow 3 [NewReno]: 0.0868988 Mbps
Flow 4 [NewReno]: 0.0937004 Mbps
Flow 5 [NewReno]: 0.0926654 Mbps
Flow 6 [NewReno]: 0.0944397 Mbps
Flow 7 [NewReno]: 0.0916303 Mbps
Flow 8 [NewReno]: 0.0934047 Mbps
Flow 9 [NewReno]: 0.0926654 Mbps
Flow 10 [Harshita]: 0.0911868 Mbps
Flow 11 [Harshita]: 0.0911868 Mbps
Flow 12 [Harshita]: 0.0876381 Mbps
Flow 13 [Harshita]: 0.0923697 Mbps
Flow 14 [Harshita]: 0.0880817 Mbps
Flow 15 [Harshita]: 0.0907432 Mbps




```
  All done!  Output plots:
  results/performance/plot_cwnd.png
  results/performance/plot_ssthresh.png
  results/performance/plot_rtt.png
  results/performance/plot_throughput.png
  results/performance/plot_dashboard.png
  results/fairness/plot_fairness.png
  results/fairness/fairness_table.txt
harshita@halcyon:~/ns-allinone-3.44/ns-3.44 $
```

10. Conclusion

This assignment implemented TcpHarshita, a new TCP CCA that enhances NewReno's slow-start with HyStart, and evaluated it comprehensively in ns-3.44.

Key experimental findings:

- HyStart exits slow start at cwnd ≈ 23 KB vs NewReno's 70 KB, preventing buffer overflow during startup.
- Peak RTT reduced from 440 ms to 300 ms (32% improvement) during the slow-start phase.
- Steady-state throughput is identical (~1.84 Mbps aggregate) — HyStart does not penalise long-term performance.
- Fairness index > 0.986 in all scenarios; the mixed scenario at 4 flows shows a real (not artefactual) difference of 0.006, confirming the per-node fix worked correctly.
- TcpHarshita is TCP-fair and backward-compatible with TcpNewReno flows in mixed deployments.